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Problem 1

a) Max 15 minute kW demand

Cust #1 - 10.84 kW

Cust #2 - 8.52 kW

Cust #3 - 11.04 kW

Cust #4 - 2.56 kW

b) Cust #1 - $\frac{\text{total energy}}{\text{hours}}$

kwh = tot energy

kwh = 116.2 kW

Avg Demand = $\frac{116.2}{4}$

$\boxed{= 29.05 \text{ kW}}$

Cust #2

$$\text{Avg Demand} = \frac{70.5}{4}$$

$$\boxed{= 17.625 \text{ kW}}$$

Cust #3

$$\text{Avg Demand} = \frac{90.26}{4}$$

$$\boxed{= 22.565 \text{ kW}}$$

Cust #4

$$\text{Avg Demand} = \frac{28.93}{4}$$

$$\boxed{= 7.23 \text{ kW}}$$

c) total kWh usage

Cust #1 - 116.2 kW

Cust #2 - 70.5 kW

Cust #3 - 90.26 kW

Cust #4 - 28.93 kW

d) Load factor

$$= \frac{\text{Avg 15-min kW demand}}{\text{Max 15-min kW demand}}$$

Cust #1 - $\frac{29.05}{116.2}$

$\boxed{= .25}$

Cust #2 - $\frac{17.625}{70.5}$

$\boxed{= .25}$

Cust #3 - $\frac{22.565}{90.26}$ $\boxed{= .25}$

Cust #4 - $\frac{7.23}{28.93}$

$= .2499$

2)

a) Max 15 min diversified demand

- 11.04 kW

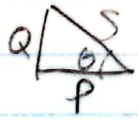
b) Max 15 min non coincident demand

= 33.12 kW

c) Utilization factor

$= \frac{\text{Max Demand}}{\text{Rated Capacity}}$

$= \frac{11.04 \text{ kW}}{25 \text{ kVA}}$



$S\cos\theta = P$

$= \frac{11.04 \text{ kW}}{25 \text{ kW}}$

25

$$\boxed{= .44} \quad UF$$

e) Load Diversity

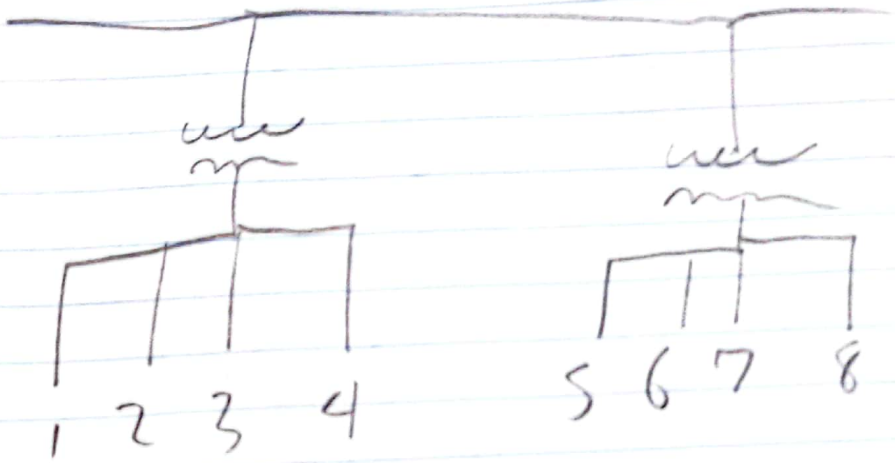
$$\frac{\text{Max Noncoincident Dem}}{\text{Max Diversified Demand}}$$

$$= \frac{33.12}{11.04}$$

$$\boxed{= 3}$$

3)

#2



a) T_1 Max kVA demand

$\boxed{= 500 \text{ kVA}}$

T_2 Max kVA demand

$\boxed{= 703 \text{ kVA}}$

b) Noncoincident max kVA demand

$T_1 = 120$ kVA $= 20 + 50 + 20 + 30$

$T_2 = 170$ kVA $= 30 + 40 + 50 + 50$

C) Load factor

$T_1 = .5$

$T_2 = .45$

$500 \cdot .9 = P$

$\frac{450\ kW}{2}$

$\frac{225\ kwh}{450} = .5$

$703 \cdot .9 = P$

$\frac{632.7}{2}$

$\frac{316.35}{703} = .45$

D) Diversity factor

$$DF = \frac{\text{Max non coincidental}}{\text{Max diversified}}$$

$T_1 = \frac{120}{50}$

$= 2.4$

$T_2 = \frac{170}{50}$

$= 3.4$

C) Load factor

$T_1 = .5$

$T_2 = .45$

$500 \cdot .9 = P$

$\frac{450\,kW}{2}$

$\frac{225\,kwh}{450} = .5$

$703 \cdot .9 = P$

$\frac{632.7}{2}$

$\frac{316.35}{703} = .45$

D) Diversity factor

$$DF = \frac{\text{Max non coincidental}}{\text{Max diversified}}$$

$$T_1 = \frac{120}{50}$$

$$\boxed{= 2.4}$$

$$T_2 = \frac{170}{50}$$

$$\boxed{= 3.4}$$

e) Suggested Transformer Rating

$T_1 = 50$ kVA $\quad \frac{50 \text{ kVA}}{.9} = 55.55$

$T_2 = 50$ kVA $\quad \frac{50 \text{ kVA}}{.9} = 55.55$

f) Utilization factor

$$\frac{\text{Max Demand}}{\text{Rated Capacity}}$$

$$\frac{50}{50} = 1$$

g) $T_1 = \text{kWh} = \text{kW Demand}\left(\frac{1}{2}\right)$

$$= \frac{500(.9)}{2}$$

$$= 225 \text{ kWh}$$

$T_2 = \frac{703}{2}(.9)$

$= 316.35$ kwh

2) Max Diversified 30 min kVA demand at Tap

$= 50$ kVA ?

Problem 3)

a) Max kW demand for each customer

Cust #1 = 3.81 kW

Cust 2 = 5.81 kW

cust 3 = 4.83 kW

cust 4 = 11.29 kW

cust 5 = 6.37 kW

cust 6 = 1.56 kW

cust 7 = 13.48 kW

cust 8 = 1.80 kW

cust 9 = 8.97 kW

cust 10 = 8.99 kW

cust 11 = 12.23 kW

cust 12 = 2.63 kW

Average kW 1 = 2.39 kW

2 1.62 kW

3 3.59 kW

4 9.26 kW

5 1.54 kW

6 .81 kW

7 4.36 kW

8 .59 kW

9 3.34 kW

10 5.42 kW

11 8.91 kW

12 1.92 kW

kwH for each customer

cust 1	9.56 kW
2	6.48 kW
3	14.76 kW
4	37.04 kW
5	6.16 kW
6	3.24 kW
7	17.44 kW
8	2.36 kW
9	13.36 kW
10	21.68 kW
11	34.44 kW
12	7.68 kW

$$\text{Load factor} = \frac{\text{Avg kw}}{\text{max demand}}$$

Cust 1 = .62

2 = .27

3 = .72

4 = .82

5 = .24

6 = .24

7 = .32

8 = .32

9 = .37

10 = .6

11 = .7

12 = .73

e) diversified demand

$T_1 = 157.44$ kW

$T_2 = 153.94$ kW

f) max non coincidental

$T_1 = 30.92$ kW

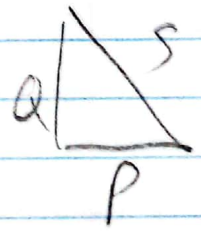
$T_2 = 49.48$ kW

g) Utilization factor

T_1 25 kVA

T_2 37.5 kVA

$pf = \cos\theta$



$S = P\cos\theta$

$$\frac{\text{Max Demand}}{\text{Rated Capacity}} = UF$$

$$T_1 = \frac{11.29}{25} = .45$$

$$T_2 \quad \frac{13.48}{37.5} = .36$$

h) $DF = \frac{\text{max non coincidental}}{\text{max diversified}}$

$$T_1 = \frac{30.92}{157.44} = .196$$

$$T_2 = \frac{49.48}{153.94} = .321$$

i) Max Diversified Demand at Node 1

$= 157.44$ kW

j) $S = VI^*$

$P = S\cos\theta$

$S = \frac{73.43}{.95}$

$= 77.29$ kVAR

$\frac{77.29}{2500}$

$= 30.96$ A

$Z = .306 + .672j\ \Omega \frac{5000}{5280}$

$= .289 + .636j\ \Omega$

$V = IZ$

$I_1 = I_2 + I_3$

$I_2 = I_3 - I_1$

$V_2 = 30.96\text{ A} \cdot (2.89 + .636j)\ \Omega$

$= 8.97 + 19.7j\ \text{V}$

$Z_2 = .306 + .672j\ \Omega \frac{7500}{5280}$

$= .435 + .954j\ \Omega$

$$I_3 = \frac{V_2}{Z_2}$$

$$= \frac{8.97 + 19.7j\ V}{.435 + .954j\ \Omega}$$

$$= 20.64 + .011j\ A$$

$$I_2 = I_3 - I_1$$

$$= (20.64 + .011j\ A) - 30.96\ A$$

$$= -10.32 + .011j\ A$$

$$S = V I^*$$

$$V_{t_1} = \frac{25\ kVA}{-10.32 - .11j\ A}$$

$= -2422.2 + 25.82, \; V$

$V_{t_1} = (-2422.2 + 25.82) - (.018\angle 40)(-10.32 + .011j)$

Low side is $\uparrow$?

$V_{t_2} =$